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Self-Service Business Intelligence solution to pharmacovigilance signal management

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Abstrakt

Hlavním cílem bakalářské práce je analýza procesů správy farmakovigilančních signálů ve společnostech farmaceutického průmyslu a následný návrh SSBI řešení, který bude možné využít k zefektivnění procesů.

Teoretická část práce představuje oblast farmakovigilance se zaměřením na procesy správy signálů a nastiňuje regulační rámec a právní požadavky EU. Praktická část poskytuje analýzu procesů, směrnic a direktiv. Na základě výsledků této analýzy práce předkládá SSBI řešení.

Hlavním přínosem této práce je vývoj integrovaného a centralizovaného nástroje pro standardizované sledování informací o nežádoucích účincích léčivých přípravků během jejich životního cyklu. Tento nástroj s různými dashboardy umožňuje sledovat metriky, usnadňuje správu zdrojů a poskytuje vizuální přehled pro management společnosti a jejich klienty. Implementací tohoto řešení lze očekávat zlepšení rozhodovacích schopností zúčastněných stran, vedoucím ke konzistentnímu procesu za účelem ochrany pacientů a veřejného zdraví.

Klíčová slova

Self Service Business Intelligence, SSBI, farmakovigilance, Power BI

Abstract

The main goal of this thesis is to analyse signal management processes within pharmaceutical industry companies and then to propose a Self-Service Business Intelligence solution which could be used to streamline the processes.

Theoretical part of the thesis introduces the science of pharmacovigilance focusing on signal management processes and outlines the EU regulatory framework and legal requirements. Practical part then contains the analysis of the processes and guidelines. Based on the results of this analysis, proposes an SSBI solution.

The main contribution of this thesis is development of an integrated and centralised tool for standardized tracking of information regarding adverse events caused by a medicine throughout their lifecycle. This tool with various dashboards enables the management to monitor compliance metrics, facilitate resource management and provide a visual overview for the companies as well as for the clients. Implementation of this solution could lead to improvement of decision-making skills of its stakeholders, leading to a consistent process void of scattering to protect patients and public health.

Keywords

Self Service Business Intelligence, SSBI, Pharmacovigilance, Power BI

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Abbreviations

ADR	adverse drug reaction
AE	adverse event
BI	Business Intelligence
EEA	European Economic Area
EMA	European Medicines Agency
ETL	Extract Transform Load
EU	European Union
GVP	Good pharmacovigilance practices
ICSR	individual case safety report
MAH	Marketing Authorisation Holder
MedDRA	Medical Dictionary for Regulatory Activities
PT	Preferred Term
PV	pharmacovigilance
SADR	suspected adverse drug reaction
SOC	System Organ Class
SSBI	Self-Service Business Intelligence

Introduction

Definition and reason for choosing the topic

The topic of the thesis is a Self-Service Business Intelligence solution to pharmacovigilance signal management which was chosen based on my interest in Business Intelligence gained during the course 4IT336 - Business Intelligence Fundamentals and work experience in a pharmacovigilance service provider to pharmaceutical companies. I believe that a technology-based Business Intelligence solution could provide more efficient approach to signal management and tracking of reactions to medicinal products.

Aims of the thesis

The main goal of this thesis is to analyse signal management processes within the pharmaceutical industry companies and then, based on the analysis, propose a Self-Service Business Intelligence solution which could be used to streamline the processes. The solution consists of various dashboards that enable the management to monitor compliance metrics and facilitate resource management. Furthermore, it provides a visual overview for the company itself as well as for the clients. Implementation of this solution leads to improvement of decision-making skills of its stakeholders, leading to a consistent process void of scattering to protect patients and public health.

Method of achieving the aims

The main goal is achieved by using theoretical knowledge gained during studies as well as the authors work experience in the field. It is backed by study of the relevant literature and articles on Business Intelligence, Pharmacovigilance and pharmacovigilance Signal Management. Furthermore, it is supported by analysis of the possible sources of signals in signal management, their structure and required metrics and lastly unified with the requirements of EU legislation defining pharmacovigilance.

Assumptions and limitations of the work

Due to the sensitive nature of the data, the main limitation is that the author is unable to use real-life data. Therefore, for the results a new, random set of data will have to be created. The second limitation could arise in the limited number and complexity of the analysed processes.

Structure of the thesis

The thesis is divided into two main parts. The first theoretical part defines pharmacovigilance, signal management, the legislation and finally business intelligence. The main contribution of this part is introduction of topics and identification of issues that need to be addressed in the practical part of the thesis. The practical part is dedicated to implementation of the SSBI solution. It starts with the introductory study and summary of requirements based on the theoretical part. Based on the study conclusions, the author provides the proposed solution.

1 Annotation of the information sources

Andrews E, Moore N. Mann's Pharmacovigilance 3rd Edition (1)

This book provides a definitive reference for the whole field of pharmacovigilance – the detection, assessment, understanding and prevention of adverse effects of medicinal products. Written by international expert editors and contributors, it acts as a reference point to anything pharmacovigilance related.

Patrick Waller, Mira Harrison-Woolrych. An Introduction to Pharmacovigilance, 2nd Edition (2)

This book provides introductory guide to the science of pharmacovigilance and is the aid to understanding the key principles and in defining the key areas of focus.

Cobert B. Cobert's Manual of Drug Safety and Pharmacovigilance (3)

Cobert's manual is an essential resource for anyone working in the field of drug safety. It discusses the theory and practices and provides essential information in step-by step form on drug safety and its regulations.

ICH Harmonised Tripartite Guideline. Clinical Safety Data Management: Definitions and Standards for Expedited Reporting E2A (4), EMA Good pharmacovigilance practices Module IX (5)

Both resources act as the main day-to-day guidance regarding the drug safety processes, their legislative requirements and are the main source of pharmacovigilance terms definitions.

Novotný O, Pour J, Slánský D. Slánský David, Business Intelligence: Jak využít bohatství ve vašich datech (6)

The publication acts as the main source of information regarding Business Intelligence from the very basic principles and explains the process of design and implementation of the final BI solution.

2 Pharmacovigilance

2.1 Definition

Pharmacovigilance, even though a relatively new discipline, has the etymological roots of the term itself in Greek word *Pharmakon* meaning medicinal substance, and Latin – *Vigilia*, meaning “to keep watch”. (7)

In other words also known as drug safety, the science of **pharmacovigilance** as defined by the World Health Organization, is “*The science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other drug related problem.*” (8)

Alternatively, the European Commission defines **pharmacovigilance** as the “*Process and science of monitoring the safety of medicines and taking action to reduce the risks and increase the benefits of medicines.*” (9)

2.2 Origin

For thousands of years, medical remedies have been recognized by the mankind for their benefits, their side effects and eventually, for their potential dangers. Some of the first recorded instances of warnings that medicines may possibly cause harm date back to ancient times together with suggested ways how to avoid them or suggest proper uses. In the centuries BC Homer, Ovid and Horace refer to the effects of medicinal plants as both beneficial and harmful. (1)

The rise of pharmacovigilance however, started on January 29th, 1848 with a death of a young girl from the north of England after receiving chloroform anaesthetic before removal of an infected toenail. Although investigated, at that time it was impossible to identify what had killed her. Chloroform, then regarded and as a safer and more powerful narcotic than ether, was introduced in clinical practice and has resulted in other deaths. To investigate the tragic and unfortunate events, the *Lancet Journal* established a commission, asked to report any deaths connected to anaesthesia and published the results in 1893. Unbeknownst to the authors at that time, they managed to create the very first adverse reaction reporting scheme together with a pharmacovigilance report. (7)

The story of the modern pharmacovigilance then follows multiple serious drug scandals and eventually culminates in the single biggest change to regulation of drugs worldwide after the 1961-1962 tragedy of Thalidomide. At that time, the drug had been welcomed and unjustifiably regarded as a safe and effective treatment of nausea and vomiting in early pregnancy. This medicine was tested on animals and found to be safe; however, the tests did not examine the effects of the drug during pregnancy and tragically, with little to no regulation of medicines outside the US, the drug also proved to cause major birth defects.

In an estimated 10 000 children worldwide, it caused a variety of birth defects but particularly limb defects known as phocomelia. As phocomelia was otherwise a very rare congenital abnormality, a major increase in its incidence did not go unnoticed in Germany, where the drug was first marketed. The cause however, initially thought to be environmental. Finally, after reports from Australia were published, again in the Lancet Journal, the problem was recognised, and the drug withdrawn from sale. Still, more than 100 years after publication of the adverse effects of chloroform, it was effectively the only mechanism for drawing attention to side effects of drugs.(10) (1) (2)

Even though we will never be able to predict and prevent all the harms that may be caused by medicines, we are now able to limit the damage caused to much smaller numbers. Had the thalidomide tragedy occurred today, we would expect to be able to identify an association between the drug and the outcome after the occurrence of less than 10 cases. (2)

2.3 Aims of pharmacovigilance

The drug safety in the past used to be reactive and the process of pharmacovigilance has often been considered to start when a drug was first authorised for its use in ordinary practice. However, after the events in 1960's the focus shifted to proactive and nowadays, pharmacovigilance is commonly considered to include all safety-related activities. After development of new regulations, it is now mandatory to perform safety surveillance of drugs before they are released for wide use and also continue after. Such post marketing pharmacovigilance activities include: *“reporting requirements, collection of information into reviewable databases, and establishment of pregnancy registries.”* (1) After all, *“The ultimate purpose of pharmacovigilance is to minimise, in practice, the potential for harm that is associated with all active medicines.”* (2)

The World Health Organization states, that pharmacovigilance *“aims to enhance patient care and patient safety and to support public health programmes by providing reliable, balanced information for the effective assessment of the benefit-risk profile of medicines and vaccines.”* (8)

To sum up, the *“aim of pharmacovigilance is to protect public health by identifying, evaluating and minimizing safety issues to ensure that the overall benefits of medicines outweigh the risks.”* (11)

Absolute safety is an unattainable goal so the aim is to use medicines with an acceptable level of safety by which it is meant that there is a low probability of harm which, in the context of the disease being treated and the expected benefits of the drug, can be considered acceptable. (2) Ultimately, the goal is to always protect the patients, communicate and to make the knowledge accessible to the relevant professionals in order to minimise risk.

2.4 Basic concepts

The fundamental concepts of pharmacovigilance are opposites: benefit and risk, safety and harm. Let us now look closely at the terms and definitions.

2.4.1 Adverse Event (AE)

“Any untoward medical occurrence in a patient or clinical investigation subject administered a pharmaceutical product and which does not necessarily have to have a causal relationship with this treatment.” (4)

“Any unfavourable and unintended sign (including an abnormal laboratory finding, for example), symptom, or disease temporally associated with the use of a medicinal product, whether or not considered related to the medicinal product.” (4)

An **adverse event** (AE) therefore is an undesirable occurrence that occurs in the context of drug treatment, but which may or may not be related to the medicine. The term is correctly used only in the context of systematic data collection with no element of judgement. The AE does not imply the medicine causing the reaction. On the other hand, if the unintended effect occurs after a medicine has been given within the normal range and is attributable and generally accepted as related to the medicine, we are talking about adverse drug reaction defined below. (2)

2.4.2 Adverse Drug Reaction (ADR)

Adverse reactions are defined according to the stage of the medicinal product's life cycle.

If the medicinal product has not yet been marketed, the definition is as follows:

“All noxious and unintended responses to a medicinal product related to any dose should be considered adverse drug reactions.” (4)

For the marketed product:

“A response to a drug which is noxious and unintended and which occurs at doses normally used in man for prophylaxis, diagnosis, or therapy of disease or for modification of physiological function.” (4)

Both definitions define **adverse reactions** as any “noxious and unintended responses” however, in the pre-marketing setting the responses are directly caused by and related to any dose, while for the marketed product, they occur at the established routine dosages, during normal use. (3)

2.4.2.1 Suspected Adverse Drug Reaction (SADR)

Suspected adverse drug reaction is such a direct response to taking a drug for which there is a reasonable possibility, that the product caused the response. (4)

The need for treating the most serious drug reactions as “suspected”, rather than “confirmed” is due to the serious nature of such reactions and need of additional evidence to be confirmed. Such evidence could be obtained through a medical testing protocol known as “dechallenge and rechallenge”. The patient would need to take the drug, have the reaction, recover after withdrawal of the drug and then, take the drug again to confirm the reaction. (12)

2.4.2.2 Unexpected Adverse Drug Reaction

Unexpected adverse reaction refers to a reaction that is not listed in the product leaflet:

“An adverse reaction, the nature or severity of which is not consistent with the applicable product information (e.g., Investigator's Brochure for an unapproved investigational medicinal product).” (4)

2.4.2.3 Serious Adverse Drug Reaction

*“A **serious adverse event** (experience) **or reaction** is any untoward medical occurrence that at any dose:*

- results in death,*
- is life-threatening,*
- requires inpatient hospitalisation or prolongation of existing hospitalisation,*
- results in persistent or significant disability/incapacity, or*
- is a congenital anomaly/birth defect.” (4)*

2.4.3 Individual Case Study Report (ICSR)

*“**Individual case safety reports** shall be used for reporting to the EudraVigilance database suspected adverse reactions to a medicinal product that occur in a single patient at a specific point in time.” (13)*

“There are four minimum criteria required for ICSRs validation:

- one or more identifiable reporter*
- one single identifiable patient*
- one or more suspected substance/medicinal product*
- one or more suspected adverse reaction.” (5)*

3 Signal management

All drugs are capable of producing side effect or adverse effect. Due to the nature of limitations of clinical trials, not all adverse effects can be disclosed during that time. Such limitations can relate to small sample sizes where it is difficult to detect of rare adverse events, limited time to detect long term adverse effects or inability to test drug interactions with other drugs or food. These drawbacks are overcome by post marketing surveillance. Therefore, one of the pillars of pharmacovigilance, is **signal management** defined as:

“A set of activities performed to determine whether, based on an examination of individual case safety reports (ICSRs), aggregated data from active surveillance systems or studies, scientific literature information or other data sources, there are new risks associated with an active substance or a medicinal product or whether known risks have changed, as well as any related recommendations, decisions, communications and tracking.” (5)

The process of **signal management** itself includes activities of:

- signal detection,
- signal validation,
- signal evaluation,
- signal prioritisation,
- signal assessment,
- recommendation for actions.

The process must be tracked and is completed to identify any new risks, or risks associated with a drug that have changed. (14)

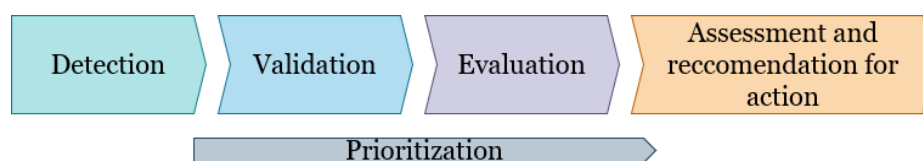


Figure 1 Stages of signal management (Source: author)

3.1 Safety signals

The European Medicines Agency defines **safety signal** as an *“information on a new or known adverse event that is potentially caused by a medicine and that warrants further investigation. Signals are generated from several sources such as spontaneous reports, clinical studies and the scientific literature” (15)*

The Council for International Organizations of Medical Sciences, Working Group VIII, defines a **signal** as *“information that arises from one or multiple sources (including*

observations and experiments), which suggest a new potentially causal association, or a new aspect of a known association, between an intervention and an event or set of related events, either adverse or beneficial, that is judged to be of sufficient likelihood to justify verificatory action.” (16)

A signal often relates to all medicinal products containing the same active substance, including combination products. Certain signals may only be relevant for a particular medicinal product or in a specific indication, strength, pharmaceutical form or route of administration whereas some signals may apply to a whole class of medicinal products. (5)

Based on the definitions, the mere presence of a **signal** does not imply causal relationship for the reported adverse event. The cause is to be assessed and determined in the process of signal management. Meyboom et al describes: “*a signal is more broadly defined as a set of data constituting a hypothesis that is relevant to the rational and safe use of a drug in humans. A signal consists of a hypothesis, together with data and arguments.*” (17) The description, rather than a definition emphasizes the fact, that signal detection is not an automatic process and requires sound clinical judgement. **Signal** does not always have to be a safety concern and on the other hand, can turn out to be a beneficial effect of a drug as well.

3.1.1 Sources of signals

The wide variety of sources where new safety signals can be detected includes all scientific information concerning the use of medicinal products including their quality, non-clinical, clinical, pharmacovigilance and pharmacoepidemiologic data. The specific sources of signals include:

- spontaneous ADR reporting systems,
- active surveillance systems,
- non-interventional studies,
- clinical studies,
- medical and scientific literature. (5) (18)

The quantitative methods on big data, which can act as another source of potential signals, can occur during expert review of aggregated data and by performing of statistical analyses in large databases.

3.2 Signal detection

Process of **signal detection**, which is a process of identification of signals from various data sources, *should follow a methodology which takes into account the nature of data and the characteristics as well as the type of medicinal product concerned. Data from all appropriate sources should be considered and clinical judgement should always be applied. It may involve a review of ICSRs, statistical analyses, or a combination of both,*

depending on the size of the data set. When it is not relevant or feasible to assess each individual case, assessment of aggregated data should be considered. (5)

The followed methodology may vary depending on the type of the concerned medicinal product, but it always needs to be defined and tracked.

The initial results of the **signal detection** process are referred to as “Potential signals”. In theory, only ADRs that are not described in the safety information of the product should be considered however an already known association may give rise to new signals if the frequency or the fatality of the event has changed. This method in its nature can create false positives and therefore a triage decision whether the potential signal is worth further follow up should be made and documented.

3.3 Signal prioritisation

One of the key elements of the signal management process is timely recognition of signals with important impacts on public health. Such signals require urgent attention and need to be prioritised for further assessment promptly. It should be a continuous process performed during the whole process.

3.4 Signal validation

Potential signals should undergo further validation considering the initial source of the information. It should take into account the clinical relevance, interactions, seriousness and additionally where required by legislation, it should also consider information on relevant case reports captured by system for reporting and evaluating suspected adverse drug reactions. In simple words it is a decision if further analysis is necessary.

Validation can result in the rejection or confirmation of a potential signal. The decision again, must be documented.

3.5 Signal evaluation

Because mere presence of a signal does not imply causal relationship between the medicine and the reported event, the signals undergo evaluation to determine the impact on public health.

When evaluating a signal the key areas to focus are:

- Causality - was the drug responsible?
- Frequency
- Seriousness

- Other clinical implications – even if the ADR is not serious, are there any other effects for patients and healthcare systems (e.g. inability to work, need for extensive investigations)?
- Preventability (2)

3.6 Assessment and recommendation for action

During assessment of the signal a conclusion is reached about the causal relationship after which the recommendation for action may suggest need for further investigations, might suggest periodic review of the signal, an update of the product information or might require immediate action such as batch recall or suspension of the whole medicinal product.

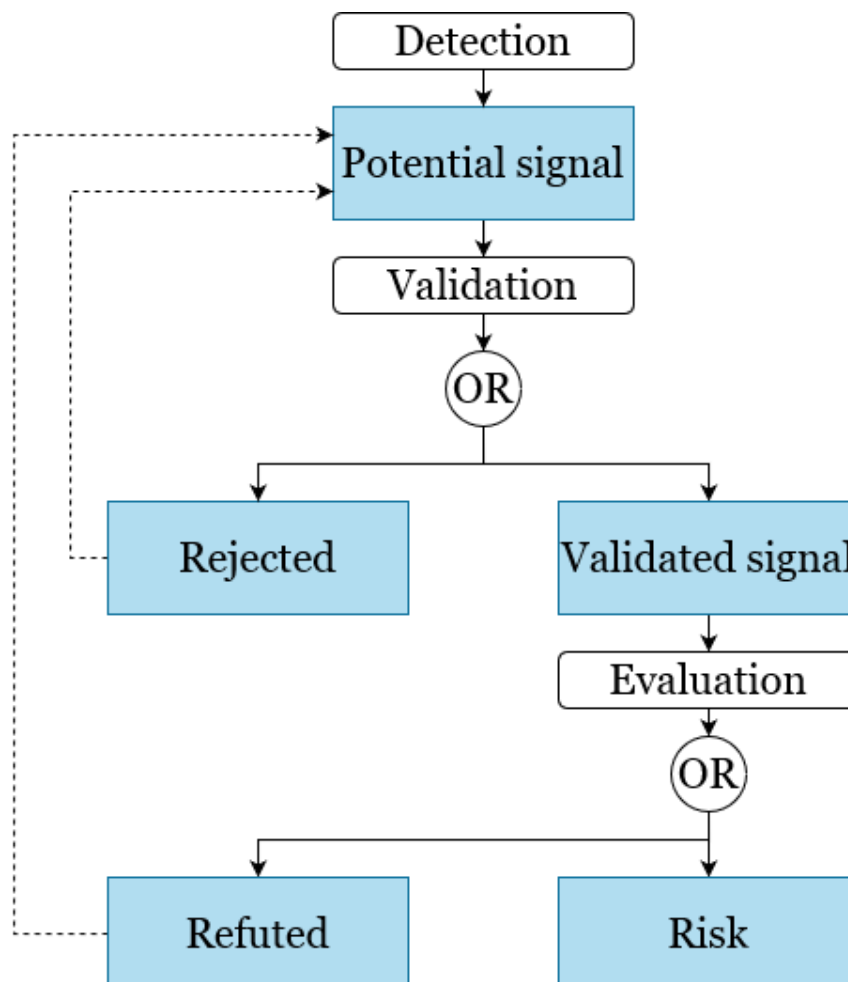


Figure 2 Simplified lifecycle of a safety signal (Source: author)

4 Legislation

Collection, accurate processing and evaluation of data from all around the world is the key element of the science of pharmacovigilance. Today, the EU regulation covers the whole lifecycle of medicinal products from their development, manufacturing through clinical trials to marketing authorisation and the following pharmacovigilance. The legislation has two broad aims. One of which is the protection of public health and the second is the creation of a single market for pharmaceuticals. (2) It is important to distinguish between EU institutions which pass regulations and set the policy frameworks and national institutions responsible for implementing the mentioned policies.

The European commission is an institution supposed to represent European interests and is the key element in initiation of legislation. The Council of Ministers is the institution that represents the Member States and together with European Parliament passes legislative acts. The European Parliament, representing the people of Europe, has a limited role regarding the practical implementation of policies, however it plays an important role regarding budgetary oversight and control of the European Medicines Agency.

The second level of national institutions is responsible for implementation of EU directives into national law. This level consists of the national competent authorities, pharmaceutical companies, and other stakeholders. Additionally, it also consists of the European Medicines Agency (EMA) (19)

4.1 Czech Republic

The State Institute for Drug Control is a Czech administration agency established by the Act no. 79/1997 Coll, falling under the direct control of the Ministry of Health. It regulates the safe production of pharmaceuticals in the Czech Republic, is responsible for the surveillance of advertising and marketing medicines and medical devices. (20)

The legal frameworks binding pharmacovigilance in the Czech Republic are:

- **Act No 378/2007 Coll.**, on pharmaceuticals and on amendments to some related acts (“Act on Pharmaceuticals”) – came into force on December 31. 2007. Title V, section 90 - 96.
- **Decree No 228/2008 Coll.**, on the marketing authorisation of medicinal products, as amended; section 14-17 (21)

4.2 European Union

Pharmacovigilance legislation was developed on the observation that adverse drug reactions caused around 197,000 deaths per year in the EU. It came into effect in July 2012

and was the biggest change to the regulation of human medicines in the European Union since 1995.

The process of review of the European system of safety monitoring started with adoption of a **Directive 2010/84/EU** and **Regulation (EU) No 1235/2010** by the European Parliament and Council of Ministers in December 2010. This legislation was then accompanied by the implementing regulation, a legally binding act published by the European Commission in June 2012 that provides details on the operational aspects for the legislation: Commission Implementing Regulation No 520/2012 of 19 June 2012. (22)

Signal management itself is defined in article 107h of **Directive 2001/83/EC** as amended, article 28a of **Regulation (EC) No 726/2004** as amended and chapter III of **Commission Implementing Regulation (EU) No 520/2012** as amended. (23)

4.2.1 European Medicines Agency

“The European Medicines Agency (EMA) is a decentralised agency of the European Union (EU) responsible for the scientific evaluation, supervision and safety monitoring of medicines in the EU.” (24) The **EMA** coordinates network of some 4000 pharmacovigilance experts and publishes guidelines which inform on safety, efficacy and quality of medicinal products. When pharmaceutical companies aim to place any new drug onto the European market, it acts as the point of direct contact and carries out the evaluation of the quality, efficacy and safety of the product. (25) Its mission is to protect human and animal health through: helping to facilitate development and access to medicines; evaluation of applications for marketing authorisations; monitoring of safety of the medicines across their whole lifecycle; and provision of reliable information on human and veterinary medicines in lay language. (26)

ADR reports, as mentioned in chapter 2, may be generated by individual doctors, nurses, other healthcare professionals, pharmacists, the patients themselves, their relatives, friends or any other person. The route of submission can be through national regulatory authorities, or regional centres or to the pharmaceutical companies or their service providers. The pharmaceutical companies are obligated to maintain a safety database allowing for reporting that is timely and in the correct format for the national authorities and EMA. (25)

4.2.2 EudraVigilance

Since 2001, EMA manages an internet-based information system, the **EudraVigilance** database. It is a data-processing network and management system, where reports of suspected adverse reactions are collected as such collection of ADRs occurring in the EU is legally required. It is used for electronic exchange among the EMA, member states' national health authorities and Marketing authorisation holders for detection of possible safety signals and their continuous monitoring and evaluation.

“EudraVigilance is the system for managing and analysing information on suspected adverse reactions to medicines which have been authorised or being studied in clinical trials in the European Economic Area (EEA). The European Medicines Agency (EMA)

operates the system on behalf of the European Union (EU) medicines regulatory network.” (27)

Since its introduction, the database currently holds total 904,559 of medicinal product submissions, over 18.6 million individual case safety reports (ICSRs) referring to over 10.5 million individual cases and is one of the largest pharmacovigilance databases in the world. In 2020, 1,821,211 ICSRs were collected and managed. The numbers presented below refer to the ADR reports received in the post-authorisation module (EVPM) (28)

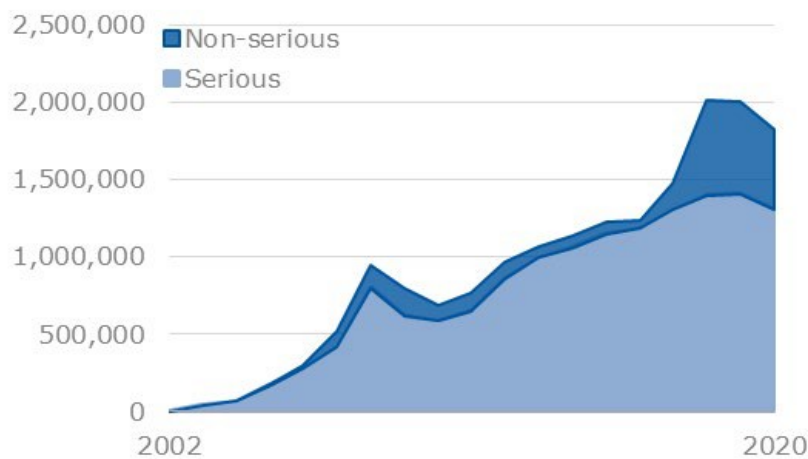


Figure 3 Number of ADR reports processed per year in EVPM. (Source: European Medicines Agency, 2021)

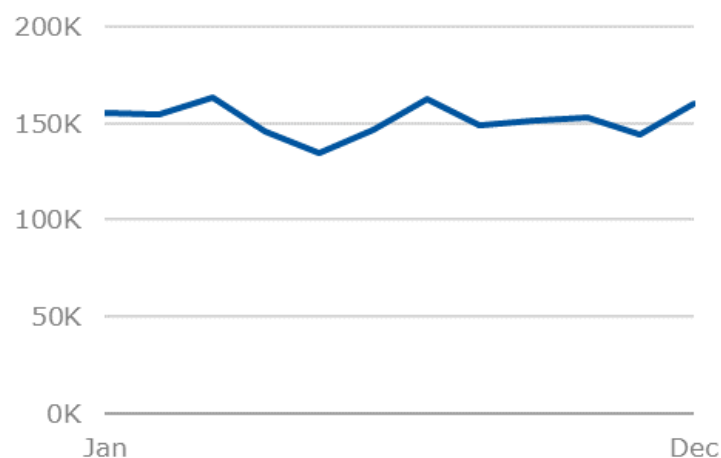


Figure 4 Number of ADR reports processed per month in EVPM in 2020. (Source: European Medicines Agency, 2021)

4.2.3 Good pharmacovigilance practices (GVP)

Medicinal legislation is supported by guidance giving practical advice on how to comply with the law. Following guidelines is generally a good practice, but it may not always be possible or appropriate. Guidelines are much more easily amended than legislation and tend to increase in size as issues of interpretation are addressed. (2)

Signal management, as a crucial part of pharmacovigilance, follows and is defined by guidelines of various degrees and the signal detection process, in each involved organisation, should be documented. In July 2012, when the new legislation for pharmacovigilance was established, the European Union developed a set of specifications for its management. **The Good pharmacovigilance practices** *“are a set of measures drawn up to facilitate the performance of pharmacovigilance in the European Union”*. (29) The guidance of signal detection may be found in the EMA Guideline on Good Pharmacovigilance Practices, GVP Module IX and in its Addendum I – which describes methodological aspects of signal detection from spontaneous reports of suspected adverse reactions.

5 Business intelligence

5.1 Early history

The term **Business Intelligence** was first used for the first time in year 1958 by visionary IBM technology scientist Hans-Peter Luhn in his research paper called “A Business Intelligence System”. He envisioned an intelligence system focused on computer-automated document processing developed to disseminate information to various sections of any industrial, scientific or government organization. Such system would be able of auto-abstraction, auto-encoding, auto-indexing, selective dissemination of information and information retrieval. (30)

Luhn was far ahead of his time, saying his business intelligence system was “*to supply suitable information to support specific activities carried out by individuals, groups, departments, divisions, or even larger units,*” and enable “*discovering information which has a bearing on a given situation.*” He was also aware of the need of collaboration and communication regarding business information. The system would “*channel a given item of information to those who need to know it*” and find co-workers “*whose interests or activities coincide most closely with a given situation.*” (31)

Over the years since 1980’s, Business Intelligence in the way we know it today underwent a series of innovations starting with financial planning languages, Executive Information Systems, based on multidimensional storage of data and processing, the data warehouse, data marts, online analytical processing and data mining. Today, it is all under one roof in a single system. (6) (32)

5.2 Definition

Even though the term had been used decades before, in 1989 Howard Dressner, researcher at Gartner Group defined the business intelligence “*as an umbrella term to describe concepts and methods to improve business decision making by using fact-based support.*” The term was widely accepted by professionals, vendors and general managers and managed to replace terms like executive information systems. (32)

Forrester Research defines **Business Intelligence (BI)** as “*a set of methodologies, processes, architectures, and technologies that transform raw data into meaningful and useful information.*” Further allowing “*business users to make informed business decisions with real-time data that can put a company ahead of its competitors.*” (33)

BI therefore covers methodologies, practices, the infrastructure, and applications that allow its users to access, analyse and transform data into useful information to the right people in the right place at the right time to support their decision making.

5.3 Basic concepts

To process and analyse the big amounts of data, it needs to be easily accessible and most importantly quickly accessible from multiple views, for which traditional relational modelling is not sufficient.

5.3.1 Multidimensional modelling

“Multidimensional models categorize data as being either facts with associated numerical measures, or as being dimensions that characterize the facts and are mostly textual.” (34)

The **multidimensional data model**, representing real-world entities, is composed of logical cubes, measures, dimensions, hierarchies, levels and attributes. Analysts then know based on which business measures they are interested in examining, which dimensions and attributes make the data meaningful, and how the dimensions of their business are organized into levels and hierarchies. (35)

5.3.1.1 Dimensions

Dimensions contain a set of unique values that identify and categorize data (35) and are used for two purposes:

- selection of data,
- grouping of data at a desired level of detail.

To achieve this, a **dimension** is organized into hierarchies composed of numerous levels, each of which represents a level of detail at a given granularity. The instances of such hierarchies are typically called dimension values. Each value belongs to a particular level. (34)

5.3.1.2 Facts

Facts are the objects that represent the subject, that is to be analysed. In most multidimensional data models, the **facts** are implicitly defined by their combination of dimension values.

It is commonplace to distinguish among three kinds of facts, that can be found together:

- event facts – captures unique instances of events e.g., a particular sale of a given product in a given store at a given time,
- state facts - models the state of a given process at a given point in time e.g., inventory levels
- cumulative snapshot facts - used to handle information about a process up to a certain point in time. E.g., the total sales in the year to date (34)

5.3.1.3 Measures

In a multidimensional database, measures generally represent the properties of the chosen facts that the users study. Because **measures** are typically multidimensional, a single value in a measure must be qualified by a member of each dimension to be meaningful. (35)

A **measure** has two components:

- numerical property of a fact, e.g., the sales price or profit,
- formula, e.g. a simple aggregate function such as SUM (34)

5.3.1.4 Data Cubes

Data cubes are the source of multidimensionality. Even though the name implies 3 dimensions, a data cube can have any number of dimensions while an addition of a dimension is easily handled. A collection of related cubes is commonly referred to as a multidimensional database or a multidimensional data warehouse. (34)

The cubes provide a means of organization for measures that have the same shape – meaning having the exact same dimensions. Measures in the same cube have the same relationships to other logical objects and can easily be analysed and displayed together. The cells of the cube are populated by the mentioned measures with collected facts and organized by dimensions. (35)

5.3.2 Implementation of the multidimensional model

Multidimensional models can be implemented using a special binary database or relational database described below.

The relational implementation can be of a star scheme or a snowflake scheme. Both of these schemes are a convention of organizing the data with a fact table in the centre containing the data to be analysed. The fact table is connected with the dimension tables using their primary keys and store textual data in measures. Dimension tables in real life use contain great quantities of attributes.

The star scheme resembles a star in appearance and its main advantage is its short response time while viewing dimensions. The disadvantage is its low change flexibility in the structure of elements. On the other hand, the snowflake scheme divides dimension tables into individual levels of hierarchy. The main advantage of such solution is reduction of data redundancy which also results in simpler implementation of changes in data structure. (6)

The juxtaposition of the schemes can be seen below:

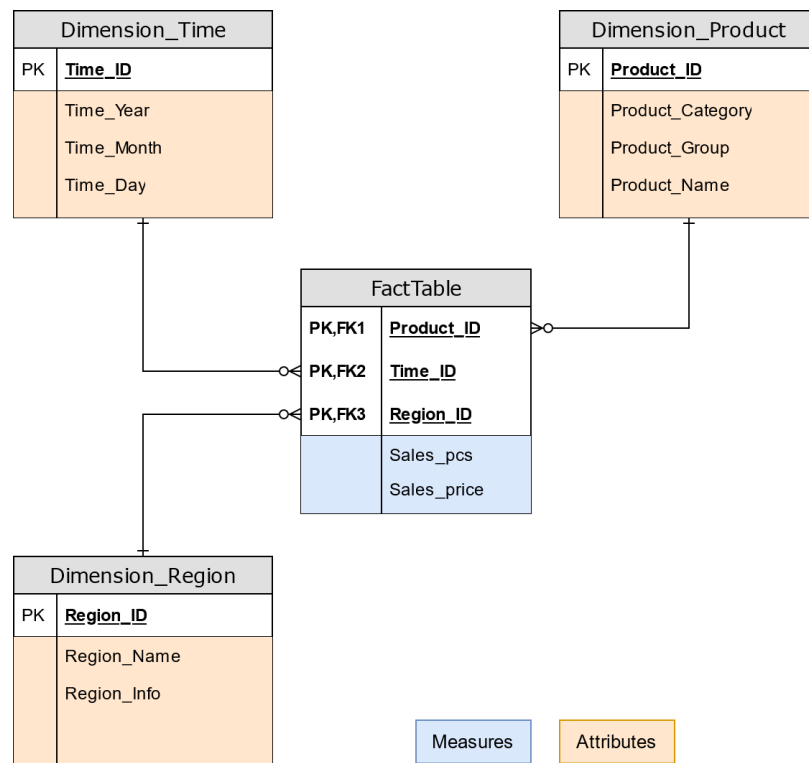


Figure 5 Star scheme (Source: author, Adapted from: Novotný et.al., 2005)

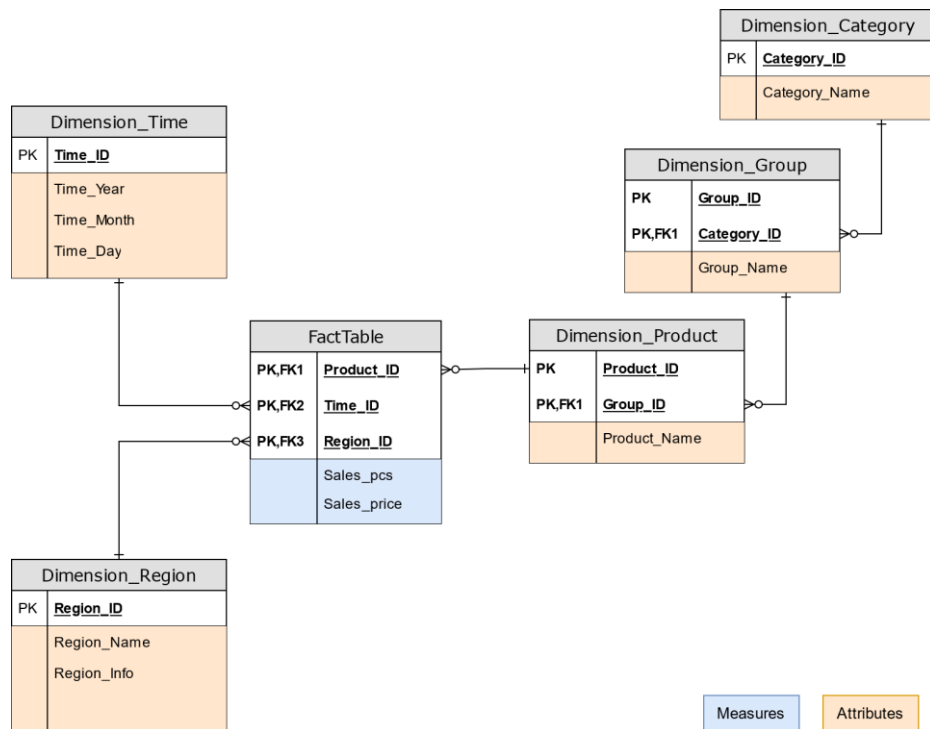


Figure 6 Snowflake scheme (Source: author, Adapted from: Novotný et.al., 2005)

5.4 Components of BI

Over the years the architecture of individual components of BI has been evolving but generally, each solution depends on requirements and resources of the sponsor. However, it can be divided into the following components:

Data Staging Area

Data Staging Area is a temporary storage of data extracted from the production databases with aim to ensure their preparation at the necessary quality. (36)

Operational Data Store

The primary objective is to gather data together from multiple primary source systems on as close to a real-time basis as possible to enable specific business processing or operational reporting before the data can be loaded and integrated further and to minimize the impact of sources. (37)

Data Warehouse

Gartner defines **Data Warehouse** as “a storage architecture designed to hold data extracted from transaction systems, operational data stores and external sources. The warehouse contains data arranged into abstracted subject areas with time-variant versions of the same records, with an appropriate level of data grain or detail to make it useful across two or more different types of analyses.” (38)

Data Mart

Definition of **Data Mart** is very similar to the one of Data Warehouse however, “A data mart contains similarly time-variant and subject-oriented data, but with relationships implying dimensional use of data wherein facts are distinctly separate from dimension data, thus making them more appropriate for single categories of analysis.” (38) Therefore Data Mart is focused on a single subject and designed for a limited circle of users.

Extract Transform Load (ETL/ELT)

ETL, commonly referred to as a data pump, is one of the most important components of the entire Business Intelligence field. Its task is to extract data from the source systems, configure and transfer them to the desired form and load them to designated data structures of the data warehouse.

The tool is used to transfer data between at least to databases or data files and it usually works in batch mode in certain intervals. The ETL data transformations are to most labour, time and financially demanding tasks of the whole process without which the BI solution would not be successful. (36)

Analytic applications

These applications are type of client applications designed to provide information to business managers to monitor company processes and help fulfil the organisational goals. They allow its users to access specific data, perform analysis and observe trends. They contain graphical user interface which makes them easy to use and therefore have high informative values. (36) Such applications can be customised and specialised or on the other hand can be in form of commonly available office software such as MS Excel with PowerPivot extension, Tableau Public and Power BI.

Reporting

Reporting is one of the main outputs of the BI solution. It is the layer that presents the results of performed analyses on the stored data in preferably, graphical form of tables and graphs, then grouped to reports or to managerial dashboards. It supports the organisation across all levels to take informed decisions. (36)

5.5 Self-service Business Intelligence

Self-service BI tools provide a comprehensive and intuitive interface that makes interacting with data more approachable for those who don't have a technical background. It puts the power of data into the hands of the user, allowing to make quicker, more informed decisions and enables exploration of data. (39)

"The self-service BI model is designed to enable people from across the organization to generate reports and perform analytical queries based on parameters they themselves define. To accomplish this, individuals must apply their knowledge of the information available and where it is stored to the process that decision makers within the organization will use to draw conclusions from it." (40)

5.5.1 Power BI

In 2021, for the fourteenth year, Gartner has recognized Microsoft as a Magic Quadrant Leader in analytics and business intelligence platforms. *"Power BI is a collection of software services, apps, and connectors that work together to turn your unrelated sources of data into coherent, visually immersive, and interactive insights. Your data may be an Excel spreadsheet, or a collection of cloud-based and on-premises hybrid data warehouses. Power BI lets you easily connect to your data sources, visualize and discover what's important, and share that with anyone or everyone you want."* (41)

6 Proposal and implementation of the Self-Service BI solution

In this chapter, the SSBI solution will be analysed, proposed and implemented.

6.1 Introductory study

The purpose of the introductory study is to provide a comprehensive analysis of the environment in which the BI solution is to be implemented. It includes definition of the main aims and benefits expected from the implementation, catalogue of the users, analysis of requirements, data source, proposal of the BI architecture and the selection of the SSBI tool. (6)

6.1.1 The main aim and benefits

Signal management information is stored across many different systems, software tools and across several functions within a pharmaceutical company. After the analysis of components of signal management in chapter 3, the main goal of the SSBI solution is to provide a tool for standardized tracking of information regarding adverse events caused by a medicine in a signal management department. The author believes that such solution could gather all relevant information into one place, streamline the processes and eventually improve decision-making skills of its stakeholders and lead to a consistent process void of scattering.

For the day-to-day users, the tool can be used to track all relevant information of adverse events and the case reports in which they are received, it can be used as a tool for detection of safety signals which can then be tracked and finally, it can provide the history of any of the safety issues received and investigated. Utilising the tool, the staff can avoid redundant work and if needed, can quickly adapt to regulatory changes.

For the manager of the department, the tool can provide an overview of the whole signal management unit and can act as a basis for creation of annual reports and can also be used to make predictions for the flowing years.

6.1.2 Catalogue of users

To fulfil the main goal, the SSBI solution will be available to Drug Safety Associates, or more experienced Officers, and to the Pharmacovigilance Physicians performing medical review of the adverse event reports. Furthermore, the tool will also be available to the managers and directors overseeing the whole process of signal management.

6.1.3 Requirements

Based on the theoretical part of the study, identification of processes that occur during a safety signal's lifecycle as well as the author's experience in company providing pharmacovigilance services to other pharmaceutical companies, to properly demonstrate the capabilities and benefits that an SSBI solution could bring to signal management, the following requirements were identified:

The solution should:

- provide general statistics about cases and events recorded,
- provide summary of all the cases and their events the company has received and that need to be reviewed,
- enable preliminary validation of the cases received – e. g. if the data provided is sufficient to reach any conclusion.

For the signal detection step the solution should provide:

- information about the events received based on the active substance involved and the reported term,
- means to identify which events can possibly trigger a signal.

Finally, for the most crucial step that needs to be documented, the solution should provide means to track all identified safety signals and act as an up-to-date signal tracking table.

6.1.4 Data source

For the solution, since the data involved is of sensitive nature, all the source data had to be created manually.

Because it would be difficult and out of the scope of the thesis to generate medically feasible and relevant data, as a framework for the source data, data extracts from the Canada Vigilance adverse reaction online database have been used. (42) It has been chosen as it *“contains information about suspected adverse reactions (also known as side effects) to health products.”* (43) that will be needed to demonstrate the features of the solution and for its structure that reflects real-life data that could eventually be used in production version of the solutions' database. Information downloaded from the Canada Vigilance database has been extracted, simplified to only contain relevant data needed for the solution, selected at random and then randomly connected. As a result of the randomness of the whole process, the information provided does not reflect any real-life situations, in no way should be used as a basis for any medical judgement and cannot be used to neither reach any conclusions, nor to give professional advice.

6.1.4.1 Medical Dictionary for Regulatory Activities (MedDRA)

To code reports of adverse event represented in the database, The Medical Dictionary for Regulatory Activities (MedDRA) terminology has been used. It *“is an internationally*

recognized set of terms used to facilitate the regulation of medical products for humans, including biopharmaceuticals, medical devices and vaccines. Developed by the International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use (ICH), MedDRA terminology is used throughout the regulatory process to enter, retrieve, analyse and present data both before and after a medical product has been authorized for sale. In addition, MedDRA enables the electronic transmission of adverse event reporting, in both pre- and post-marketing phases.” (44)

As the highest level of the terminology, the 27 MedDRA system organ classes (SOC) have been used. SOC *“is defined as the highest level of the MedDRA terminology, distinguished by anatomical or physiological system, aetiology (disease origin) or purpose. Most of these describe disorders of a specific part of the body. For example:*

- *Cardiac disorders describe heart problems*
- *Renal and urinary disorders describe kidney and bladder problems”*

Furthermore, on the lower level under the SOC, terms used to describe adverse events use the level of *“MedDRA 'preferred term' (PT), which describes a single medical concept.*

Each MedDRA system organ class has a number of MedDRA preferred terms associated with it. For example, the system organ class 'cardiac disorders' includes (among others) the following preferred terms:

- *angina*
- *cardiac arrest*
- *myocardial infarction*
- *palpitations.” (45)*

6.1.5 SSBI tool

Generally, the information that will be centralised in the SSBI solution, is stored across multiple separate databases using a wide variety of tools, the most common being MS Excel, and can also take form of paper-based systems. To provide the solution, the tool of choice is the Microsoft Power BI Desktop. It has been chosen as it is available for free as a part of the Microsoft Office portfolio, for its familiar interface and user friendliness, and for its ability to be customized for any use-case.

6.2 Dimensional modelling

Chapter dimensional modelling contains information about the fact tables and the dimensions connected to create the dimensional model.

6.2.1 Dimensional model

After the analysis of the requirements the solution should provide, the following need of fact tables has been identified:

- F_Case
- F_Event
- F_Signal

To further determine the needed dimensions, let us first explore the fact tables that are proposed for the solution.

6.2.1.1 Fact table Case

The fact table **Case** will be used to track the case reports received.

Table 1 Fact table F_Case (Source: author)

F_Case	
Attributes	Case_ID Case_Version Case_InitialReceivedDate_ID Case_LatestReceivedDate_ID ReportType_ID ReportSource_ID ReporterType_ID Gender_ID Age_ID ActiveSubstance_ID Outcome_ID
Calculated columns	Patient information complete Reporter information complete Number of events Validity Is serious case

Each case report has its own, unique ID (Case_ID) by which it is identified. Cases are also identified by a version number (Case_Version) which represents the amount of submitted follow-ups. Therefore, the initial received date (Case_InitialReceivedDate_ID) notes receipt of the version 0 of the case report and the latest received date (Case_LatestReceivedDate_ID), which can be different from the initial date, notes the receipt of the latest follow-up report.

Next in the fact table, the type of the report from which the case originates (ReportType_ID) is recorded e. g. report from an individual or a literature article. Its source (ReportSource_ID) is also recorded together with a type of reporter (ReporterType_ID) e. g. a physician or the consumer of the medicinal product.

For the consumer, or in other words the patient involved, their age (Age_ID) and gender (Gender_ID) is recorded. Furthermore, the outcome of the case (Outcome_ID) records information whether the patient has recovered or not.

Also recorded is the active substance (ActiveSubstance_ID) that is involved in the events of the case.

To make sure that the case itself is valid, several calculated columns are introduced:

- Firstly, a column to check if the information about a patient is complete:

```
Patient information complete =  
IF(OR(ISBLANK(F_Case[Gender_ID]),ISBLANK(F_Case[Age_ID])),FALSE(),TRUE())
```

- Secondly, check if the information about the reporter is complete:

```
Reporter information complete =  
IF(OR(OR(ISBLANK(F_Case[ReportType_ID]),ISBLANK(F_Case[ReportSource_ID])),ISBLANK(F_Case[ReporterType_ID])),FALSE(),TRUE())
```

- Third, a number of events reported for a case:

```
Number of events = COUNTROWS(FILTER(F_Event, F_Event[Case_ID] = F_Case[Case_ID]))
```

- Lastly a column that determines if the case is valid:

```
validity = IF(AND(AND(AND(F_Case[Patient information complete],F_Case[Reporter  
information complete]),F_Case[Number of  
events]>0),NOT(ISBLANK(F_Case[ActiveSubstance_ID]))),1,0)
```

Furthermore, the case can consist of multiple events that can either be serious or not serious. If at least one of the events is determined as serious, the whole case is considered as serious as well:

```
Is serious case =  
IF(CALCULATE(COUNTROWS(F_Event),D_Seriousness[Seriousness_Type]="Serious")>0,TRUE(),FALSE()  
)
```

6.2.1.2 Fact table Event

The fact table **Event** will be used to track the events received for each individual case.

Table 2 Dimension F_Event (Source: author)

F_Event	
Attributes	Event_ID Case_ID Code_PT_ID Seriousness_ID Causality_ID

F_Event	
Calculated columns	Active substance Latest received date Outcome Is valid Gender A B C D RoR

Each event has its own, unique ID (Event_ID) by which it is identified and can belong to one case (Case_ID). The one particular event itself, is coded in MedDRA terminology to a preferred term (Code_PT_ID). Also recorded is the seriousness of the event (Seriousness_ID) and the opinion of the reporter whether the adverse event is related to the medicinal product – the causality (Causality_ID).

To link the event to the case and get the needed information, the calculated columns are as follows:

```
Active substance =
LOOKUPVALUE(F_Case[ActiveSubstance_ID],F_Case[Case_ID],F_Event[Case_ID])
```

```
LatestReceivedDate_ID =
LOOKUPVALUE(F_Case[Case_LatestReceivedDate_ID],F_Case[Case_ID],F_Event[Case_ID])
```

```
Outcome = LOOKUPVALUE(F_Case[Outcome_ID],F_Case[Case_ID],F_Event[Case_ID])
```

```
Is Valid = LOOKUPVALUE(F_Case[Validity],F_Case[Case_ID],F_Event[Case_ID])
```

```
Gender = LOOKUPVALUE(F_Case[Gender_ID],F_Case[Case_ID],F_Event[Case_ID])
```

For the purposes of quantitative signal detection, a disproportionality statistics method of reporting odds ratio (ROR) is introduced and calculated. It takes „*the form of a ratio of the observed proportion of spontaneous ICSRs with a medicinal product that include a specific adverse event to the proportion that would be expected if no association existed between the product and the event.*“ (46)

Table 3 Variables for ROR calculation (Source: author, Adapted from: European Medicines Agency, (46))

	Event	All other events
Active Substance	A	B
All other active substances	C	D

```
A = COUNTX (
  FILTER ( F_Event, EARLIER ( F_Event[Code_PT_ID] ) = F_Event[Code_PT_ID] &&
    EARLIER(F_Event[Active substance]) = F_Event[Active substance]),
    F_Event[Event_ID]
)
```

```
B = COUNTX (
  FILTER ( F_Event, EARLIER ( F_Event[Code_PT_ID] ) <> F_Event[Code_PT_ID] &&
    EARLIER(F_Event[Active substance]) = F_Event[Active substance]),
    F_Event[Event_ID]
)
```

```
C =
COUNTX (
  FILTER ( F_Event, EARLIER ( F_Event[Code_PT_ID] ) = F_Event[Code_PT_ID] &&
    EARLIER(F_Event[Active substance]) <> F_Event[Active substance]),
    F_Event[Event_ID]
)
```

```
D = COUNTX (
  FILTER ( F_Event, EARLIER ( F_Event[Code_PT_ID] ) <> F_Event[Code_PT_ID] &&
    EARLIER(F_Event[Active substance]) <> F_Event[Active substance]),
    F_Event[Event_ID]
)
```

The ROR is then calculated as follows:

$$ROR = \frac{\frac{A}{B}}{\frac{C}{D}}$$

```
ROR = DIVIDE(DIVIDE(F_Event[A],F_Event[B]),DIVIDE(F_Event[C],F_Event[D]))
```

6.2.1.3 Fact table Signal

And lastly a fact table **Signal** will be used to track identified signals originating from the events.

Table 4 Dimension F_Signal (Source: author)

F_Signal	
Attributes	Signal_ID Event_ID DateOfDetection_ID DateOfValidation_ID DateOfClosure_ID SignalStatus_ID
Calculated columns	Date of event receipt Date of validation Date of closure Active substance Signal term Days to detect Days open

Each signal needs to have its own, unique ID (Signal_ID) by which it is identified and can belong to one event (Event_ID). For each signal, we record three dates:

- Date of detection (DateOfDetection_ID) – date when the signal was detected, that cannot be empty
- Date of validation (DateOfValidation_ID) – date when the signal has gone the step of validation, if applicable
- Date of closure (DateOfClosure_ID)– date when the signal was closed, if applicable

Finally, the status of a signal (SignalStatus_ID) is tracked.

Because there are multiple dates connected to a date dimension with only the Date of detection being in an active relationship, the rest of the dates need to be connected via inactive relationships. To show the dates in a readable format for a dashboard visualization, the calculated columns are as follows:

```
Date of event receipt =  
LOOKUPVALUE(F_Event[LatestReceivedDate_ID],F_Event[Event_ID],F_Signal[Event_ID])
```

```
Date of validation =  
LOOKUPVALUE(D_Date[Date],D_Date[Date_ID],F_Signal[DateOfValidation_ID])
```

```
Date of closure = LOOKUPVALUE(D_Date[Date],D_Date[Date_ID],F_Signal[DateOfClosure_ID])
```


To get the relevant information from the relevant event fact table for the purposes of signal tracking, the calculated columns are:

```
Active substance = LOOKUPVALUE(F_Event[Active
substance],F_Event[Event_ID],F_Signal[Event_ID])
Signal term = LOOKUPVALUE(F_Event[Code_PT_ID],F_Event[Event_ID],F_Signal[Event_ID])
```

And finally, calculated columns to calculate date differences for measures:

- The difference in days for a signal to be detected since the relevant event has been received:

```
Days to detect = DATEDIFF(LOOKUPVALUE(D_Date[Date],D_Date[Date_ID],F_Signal[Date of
event receipt]),RELATED(D_Date[Date]),DAY)
```

- The difference in days during which the signal remains open:

```
Days open = DATEDIFF(RELATED(D_Date[Date]),F_Signal[Date of Closure],DAY)
```

6.2.1.4 Dimension Date

The cases, events, and signals store different dates. To store them, the following dimension has been created:

Table 5 Dimension D_Date (Source: author)

ID	D_Date
Dimension name	Date
Content	Dates to track metrics over time
Type	Time
Source	Excel
Structure	Date
Attributes	Date_ID, Date

6.2.1.5 Dimension Report type

Each case that is received has a form of a report. In our proposed solution we are working with the following types of report:

- Literature

- Spontaneous
- Study
- Other

To store the type of the report in which a case has been received, the following dimension has been created:

Table 6 Dimension D_ReportType (Source: author)

ID	D_ReportType
Dimension name	Report type
Content	Types of a report from which a case can originate
Type	Star
Source	Excel
Structure	Report type
Attributes	ReportType_ID, ReportType_Name

6.2.1.6 Dimension Report source

Each report has its own source from which it has been received. In our proposed solution we are working with the following sources:

- Community
- Hospital
- Clinical Study
- MAH – Marketing Authorisation Holder – “*The company or other legal entity that has the authorisation to market a medicine*” (47)

To store the source of the report from which a case has been received, the following dimension has been created:

Table 7 Dimension D_ReportSource (Source: author)

ID	D_ReportSource
Dimension name	Report source
Content	Types of source of a report from which a case can originate
Type	Star
Source	Excel

ID	D_ReportSource
Structure	Report source
Attributes	ReportSource_ID, ReportSource_Name

6.2.1.7 Dimension Reporter type

Each report has its own reporter from whom it has been received. In our proposed solution we are working with the following types of reporter:

- Consumer
- Other non-health care professional
- Physician
- Nurse
- Pharmacist
- Other health care professional

To store the type of reporter form whom the report has been received, the following dimension has been created:

Table 8 Dimension D_ReporterType (Source: author)

ID	D_ReporterType
Dimension name	Reporter type
Content	Types of reporters from whom the report for a case can originate
Type	Star
Source	Excel
Structure	ReporterType
Attributes	ReporterType_ID, ReporterType_Name

6.2.1.8 Dimension Gender

Each case concerns a patient who was affected. To store the gender of the patient, the following dimension has been created:

Table 9 Dimension D_Gender (Source: author)

ID	D_Gender
Dimension name	Gender
Content	Gender of a patient
Type	Star
Source	Excel
Structure	Gender
Attributes	Gender_ID, Gender_Name

6.2.1.9 Dimension Age

Each case concerns a patient who was affected. As paediatric patients can be a significant trigger for need of an action, in our proposed solution we are working with two sets of age:

- 1 – 36 months
- 1 – 110 years

To store the age of the patient, the following dimension has been created:

Table 10 Dimension D_Age (Source: author)

ID	D_Age
Dimension name	Age
Content	Age of a patient
Type	Star
Source	Excel
Structure	Age
Attributes	Age_ID, Age_Number Age_Unit

6.2.1.10 Dimension Active substance

Each case is reported for an active substance of a medicinal product. To store the active substance of a case, the following dimension has been created:

Table 11 Dimension D_ActiveSubstance (Source: author)

ID	D_ActiveSubstance
Dimension name	Active substance
Content	Active substance of a medicinal product
Type	Star
Source	Excel
Structure	Active substance
Attributes	ActiveSubstance_ID, ActiveSubstance_Name

6.2.1.11 Dimension Outcome

Each case has its own outcome. In our proposed solution we are working with the following possible outcomes of a case:

- Unknown
- Recovered/resolved
- Recovering/resolving
- Not recovered/not resolved
- Recovered/resolved with sequelae
- Fatal

To store the outcome of the case, the following dimension has been created:

Table 12 Dimension D_Outcome (Source: author)

ID	D_Outcome
Dimension name	Outcome
Content	Outcome of the case
Type	Star
Source	Excel
Structure	Outcome
Attributes	Outcome_ID, Outcome_Type

6.2.1.12 Dimension Validity

Each case has its own determined validity and can either be valid or invalid e.g. when missing information about the reporter.

To store the validity of the case, the following dimension has been created:

Table 13 Dimension D_VValidity (Source: author)

ID	D_VValidity
Dimension name	Validity
Content	Validity of the case
Type	Star
Source	Excel
Structure	Validity
Attributes	Validity_ID, Validity_Name

6.2.1.13 Dimension Seriousness

Each event reported in a case has its own seriousness:

- Not serious
- Serious

To store the seriousness of the event, the following dimension has been created:

Table 14 Dimension D_Seriousness (Source: author)

ID	D_Seriousness
Dimension name	Seriousness
Content	Seriousness of the event
Type	Star
Source	Excel
Structure	Seriousness
Attributes	Seriousness_ID, Seriousness_Type

6.2.1.14 Dimension Causality

Each event reported in a case has its own causality. Based on this value, the causal relationship of the AE can be judged:

- Not Related
- Not Reported
- Possibly Related
- Related
- Unknown
- Unlikely Related

To store the determined causality of the event, the following dimension has been created:

Table 15 Dimension D_Causality (Source: author)

ID	D_Causality
Dimension name	Causality
Content	Causality of the event
Type	Star
Source	Excel
Structure	Causality
Attributes	Causality_ID, Causality_Type

6.2.1.15 Dimension PT

Each triggered safety signal has its own signal term for which it is coded. In our proposed solution, the first recoded level is the standardised MedDRA code terms for a Preferred Term.

To store the PT level of a signal term, the following dimension has been created:

Table 16 Dimension D_Code_PT (Source: author)

ID	D_Code_PT
Dimension name	Preferred Term
Content	PT of the coded signal term
Type	Snowflake

ID	D_Code_PT
Source	Excel
Structure	PT > SOC
Attributes	Code_PT_ID, Code_SOC_ID, Code_PT_Name

6.2.1.16 Dimension SOC

Each triggered safety signal has its own signal term for which it is coded. In our proposed solution, the second and the broadest recoded level is the standardised MedDRA code terms for a System Organ Class term.

To store the SOC level of a signal term, the following dimension has been created:

Table 17 Dimension D_Code_SOC (Source: author)

ID	D_Code_SOC
Dimension name	System Organ Class
Content	SOC of the coded signal term
Type	Snowflake
Source	Excel
Structure	PT > SOC
Attributes	Code_SOC_ID, Code_SOC_Name

6.2.1.17 Dimension Signal status

Each triggered signal can have its status defining the step of its lifecycle. In our proposed solution we are working with the following statuses:

- Potential Signal
- Rejected
- Valid
- Valid - refuted
- Monitoring
- Closed
- Closed - not validated

To store the status of a signal, the following dimension has been created:

Table 18 Dimension D_SignalStatus (Source: author)

ID	D_SignalStatus
Dimension name	Signal status
Content	Status of a safety signal
Type	Star
Source	Excel
Structure	Signal status
Attributes	SignalStatus_ID, SignalStatus_Name

The dimensional model has been automatically created by loading all the tables described above into the Power BI application. The one relationships used between tables is 1..* - one to many with many leading from the fact tables to 1 unique ID value in the dimensional tables. The final model can be seen in the figure below:

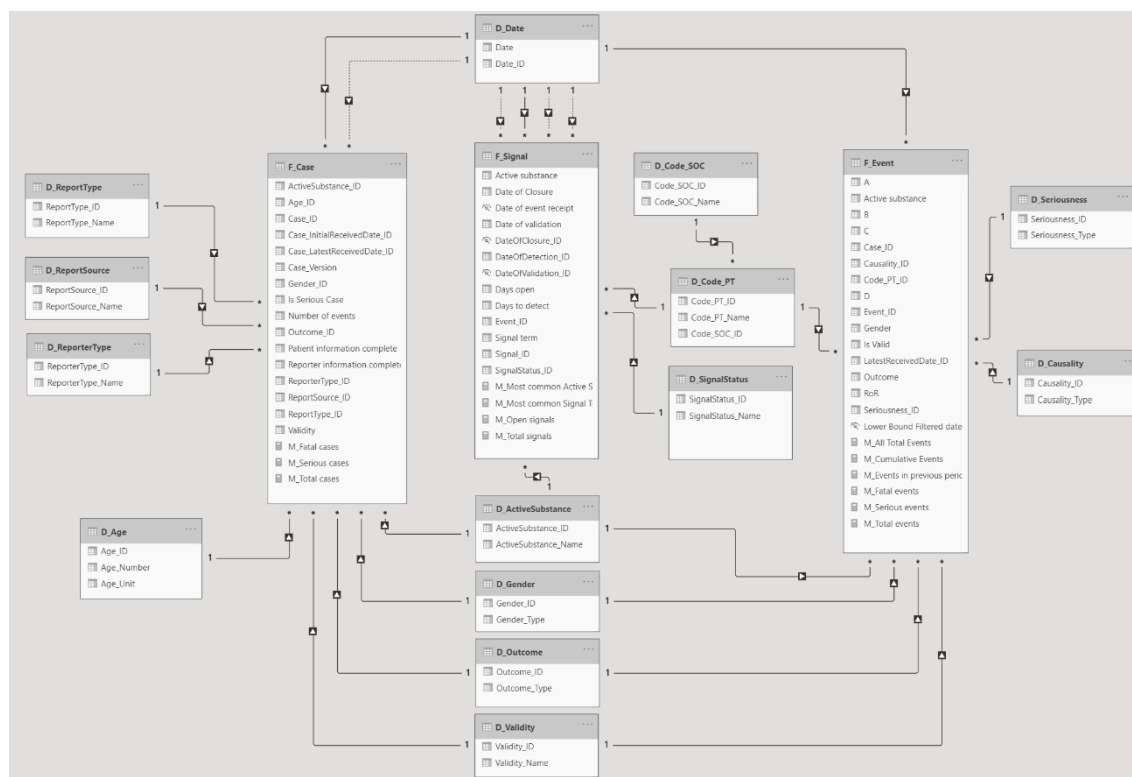


Figure 7 Dimensional model (Source: author)

6.2.2 Data metrics

Table 19 Defined metrics and their characteristics (Source: author)

ID	Name	Contents	Source	Type	Units
M_Total Cases	Total cases	Total number of reported cases	Excel	Numeric	Cases
M_Serious Cases	Serious cases	Total number of reported serious cases	Excel	Numeric	Cases
M_Fatal Cases	Fatal cases	Total number of reported fatal cases	Excel	Numeric	Cases
M_Total Events	Total events	Total number of reported events	Excel	Numeric	Events
M_Serious Events	Serious events	Total number of reported serious events	Excel	Numeric	Events
M_Fatal Events	Fatal events	Total number of reported fatal event	Excel	Numeric	Events
M_All total events	Total number of recorded events	Total number of recorded events that does not take slicers into consideration	Excel	Numeric	Events
M_Events in previous period	Events in previous period	Total number of recorded events that is before the selected period in slicers	Excel	Numeric	Events
M_Cumulative events	Cumulative events	Total number of recorded events that is before and in the selected period in slicers	Excel	Numeric	Events
M_Total signals	Total signals	Total number of tracked signals	Excel	Numeric	Signals
M_Open signals	Open signals	Total number of tracked open signals	Excel	Numeric	Signals
M_Most common Active substance	Most common active substance	Name of the most commonly reported active substance in a signal tracking table	Excel	Text	-

ID	Name	Contents	Source	Type	Units
M_Most common Signal Term	Most common signal term	Name of the most commonly reported signal term in a signal tracking table	Excel	Text	-

6.3 Dashboards

Chapter dashboard showcases the dashboards, tables and graphs created for the SSBI solution.

6.3.1 Overview

The first dashboard named Overview presents to its users, as the name itself suggests, an overview of the recorded cases and events. In the upper left corner, the data presented can be filtered by the latest received date and by the outcome. Any other filters can be applied using the visualizations themselves. Using the visualizations as filters for example, the data can be filtered for valid or invalid cases and for cases by gender. The filters can also be combined.

Information about cases contains the percentages of valid and invalid cases, number of cases per each year and together for all years, division of cases between the two genders and numbers of serious and fatal cases. Also included is the average number of events per case. Events are separated into all and serious specifically. They are further divided into categories by their reported term on SOC level and can, eventually, be decomposed to their PT level using the decomposition tree visualization in the lower right corner.

Overview

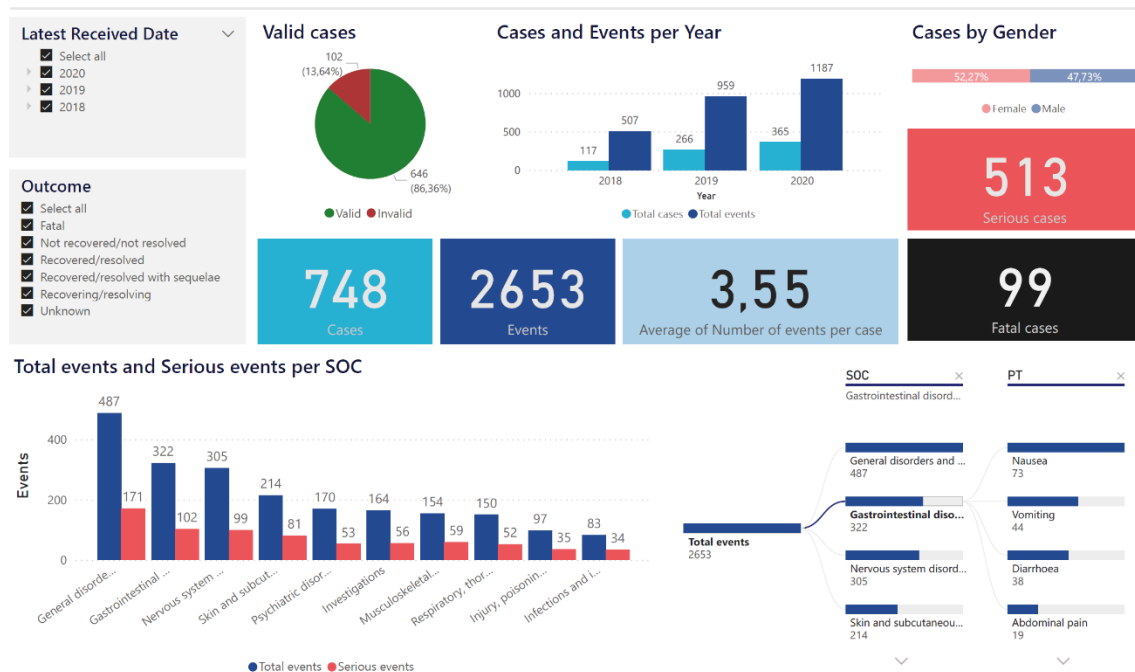


Figure 8 Overview dashboard (Source: author)

6.3.2 Signal Detection – analysis

The Signal Detection – analysis dashboard is the main screen containing information about reported adverse events needed to potentially raise a safety signal.

The data on the dashboard can be filtered by date period of the latest received date, SOC, Case validity and the determined causality of the events. The visualisations then show the total number of events received in the period prior to the selected lower bound in the date slicer and the number of events reported in the selected period.

A line graph Event receipt shows the date and number of events received in time. Furthermore, a column chart provides number of events received for active substances.

A summary table with relevant columns for each event is provided.

Signal Detection - analysis

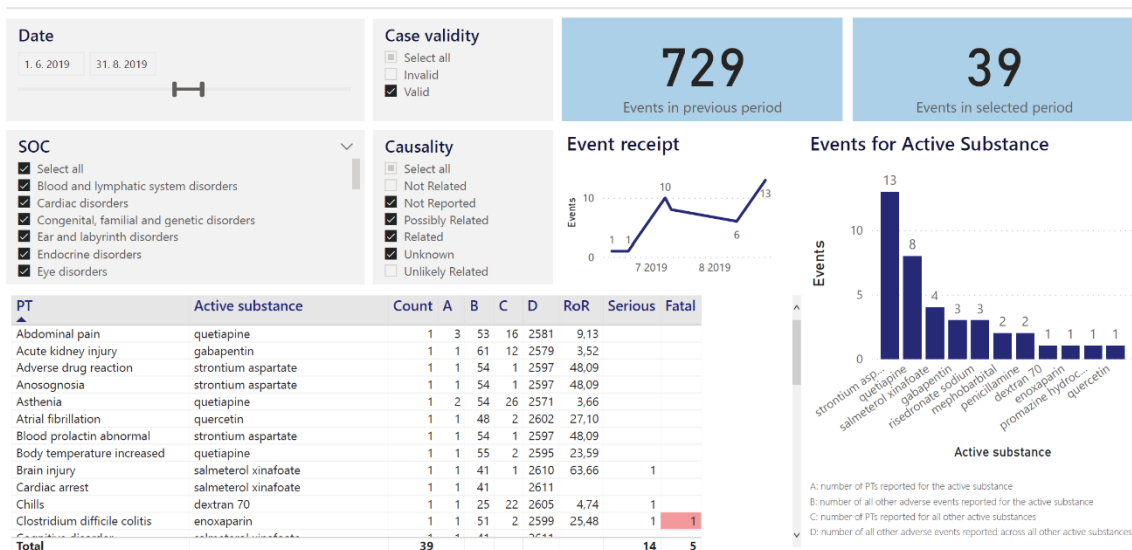


Figure 9 Signal Detection - analysis dashboard (Source: author)

6.3.3 Signal Tracking

The Signal Tracking dashboard is the main screen containing all information about signals and can act as the signal tracking table. Based on the Date of Detection slicer, the information can be filtered for any desired period. Furthermore, the dashboard can also be filtered based on the signal status.

Based on the selected criteria, the total number of signals in the period and the number of signals that are open are provided. Underneath a summary of the most common active substance and the most common signal term is provided. The users can also see the average number of days it takes for a signal to become closed, and the average number of days needed for a signal to be detected since the receipt of the relevant event.

All the information is summarised in a column chart showing numbers for each signal status category. Below, a summary signal tracking table with relevant columns for each signal is provided.

Signal Tracking

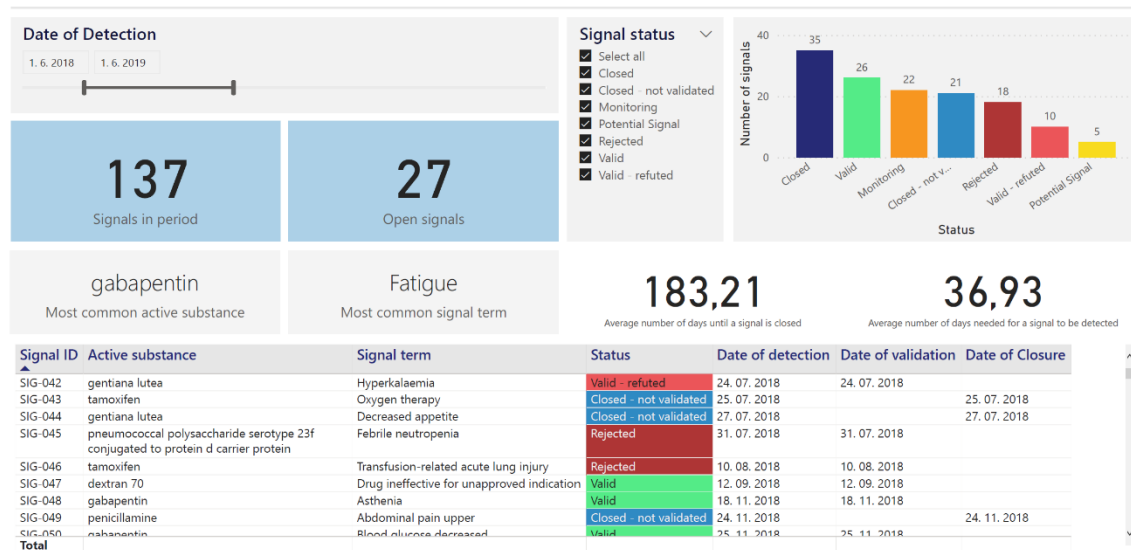


Figure 10 Signal Tracking Dashboard (Source: author)

6.4 Delivery of the SSBI solution

Based on the fact that the whole second part of the thesis, the implementation of the SSBI solution, is done for a fictional, non-existent company and acts as a proposal to pharmacovigilance companies dealing with signal management, the chapter delivery of the SSBI solution will also only deal with theoretical steps needed to implement the solution into production use.

The first step is preparation of the operational environment such as preparation of the required version of operating systems and software and deployment of the required software on servers and users' computers if needed. Furthermore, continuous data updates need to be ensured by determining the methods and timelines for refreshing of the SSBI database tables. Lastly, it is necessary to train all users of the SSBI solution. (36)

For the delivery of the proposed solution in the thesis, the first step, taken by the IT department, is ensuring that the user licenses of Microsoft Power BI are purchased and installed. On the market, there are three versions: Free, Pro and Premium. The main downside of the Free version is the lack of collaborative tools such as sharing of the visualizations and reports between users. This issue is addressed in the Pro version, which is a part of Microsoft 365 E5, the enterprise edition. (48)

The next step involves implementation of the proposed solution with the data to the cloud Power BI service to gain the possibility to share the outputs among users and to allow regular, automated updates of the database.

At the same time, the future users of the solution, the employees, need to be introduced to SSBI, what it is and what the solution can be useful for. At this step, feedback and additional requirements from the users can be obtained to further improve the proposed solution. Furthermore, to be able to use the solution in day-to-day operations, the users need to be trained and become familiar with Microsoft Power BI.

6.5 Conclusion

In this chapter, the proposal and implementation of the Self-Service Business Intelligence solution has been introduced. Starting with the introductory study including the main aims and benefits, the catalogue of users and the requirements of the solution based on the first, theoretical part of the thesis. The data source for the solution has been created and the SSBI tool has been chosen. Next step comprised of creation of the dimensional model of the solution including the fact tables, dimensions, and the relationships between them. Moreover, three dashboards have been proposed.

Conclusion

The main goal of this thesis was to analyse signal management processes within the pharmaceutical industry companies and based on the analysis, to propose a Self-Service Business Intelligence solution.

In the first, theoretical part of the thesis, the science of pharmacovigilance has been introduced including aims of pharmacovigilance and definitions of the basic concepts. Furthermore, the thesis introduced signal management as a crucial part of both pharmacovigilance and the contents of the thesis itself. Also included were the legislative requirements of pharmacovigilance in the Czech Republic and European Union. Finally, the thesis contains theory of Business Intelligence.

Based on the analysis a proposal for a Self-Service Business Intelligence solution was created. The first step was identification and creation of the necessary data which was then divided into multiple dimensions and fact tables. The prepared data was then imported into Power BI and further manipulated to create multiple measures. The solution explores and showcases options that could be used to streamline the processes. With the help of multiple dashboards, it also provides a visual overview of the recorded clinical data.

In my opinion, the main contribution of the thesis lies in its showcase of the use-cases for which an SSBI solution like the one proposed could be used, while also showing its limitations. Furthermore, the solution provides room for further extensions to other elements of signal management. For example, the cases could be extended to include their narratives which are currently not included in the database.

The main goal of the thesis, proposal of an SSBI solution has been, in my view, for the most part fulfilled. An SSBI solution has been proposed and can be helpful and used in signal management. However, even though the number of involved processes is not very broad, after the analysis they turned out to be more complex than it had been originally expected. Therefore, the solution could be improved in the future. The main area that could be addressed is a solution for tracking of the documents written during the process of signal management e. g. signal detection or validation report and periodic review report. For such documents their authors, quality and medical reviewers could be tracked together with the adverse events mentioned. Such extension could provide more in-depth view on the quality and compliance of the signal management department for the management.

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Appendices

Appendix A: Proposed SSBI solution

File “Appendix_A_PowerBI_Solution_petj22.pbix” that can be opened using Microsoft Power BI contains the proposed SSBI solution from the second, practical part of the thesis.